

Iterative MMSE Detection for Orthogonal Time Frequency Space Modulation

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Abstract—In this article, a successive interference cancellation (SIC) based minimum mean squared error (MMSE) signal detection is investigated for orthogonal time frequency space (OTFS) modulation. In this proposed detection technique, the transmit symbols will be detected on the basis of layers, and different multipath components of the same layer signal together with its interference will be coherently combined and suppressed through MMSE filtering. The performance will be further improved from an iterative operation on SIC-MMSE detection. The complexity of calculating MMSE filter coefficients can be reduced when detection is performed in the time domain due to its simpler channel structures. Simulation shows that our proposed SIC-MMSE detection can outperform maximum-ratio-combining (MRC) thanks to its inherent balances on matched filter and zero-forcing detector.

I. INTRODUCTION

With current 5G wireless technology gradually expanding its deployment, conventional modulation schemes such as OFDM will have difficulties for some high-mobility user application scenarios such as high-speed railways, autonomous vehicles, unmanned aerial vehicles (UAVs), etc., when a high Doppler shift can generate severe inter-carrier interference and degrade the overall system performance. In order to eliminate the undesired effects introduced by Doppler shift and improve the overall system performance, a recently proposed new modulation scheme named orthogonal time frequency space (OTFS) [1] serves as a promising technique that has been shown to be more robust in combating the Doppler effect experienced in doubly-selective wireless channels. In OTFS, the information symbols are modulated in the delay-Doppler domain, where the time-varying wireless channel can be regarded as almost time invariant in the delay-Doppler domain, within a stationary region. Additionally, OTFS modulates information symbol over two dimensional (2D) basis function that spreads across the entire time-frequency space, which in return can potentially achieve full time-frequency diversity [2] [3] [4]. One drawback for OTFS is that it cannot adopt a simple single-tap equalizer for the receive signals. In regards to this issue, tailored detection methods are thus required to be designed on the receiver side in order to unleash its full potentials. Several signal detection techniques have been addressed in the literature. To name a few, non-linear receivers such as the maximum a posteriori (MAP) detection in [5] can achieve the optimal performance, but at a cost of high

complexity which increases exponentially with OTFS frame length. Other message passing based detections such as the message passing algorithm (MPA) in [2] has a relatively lower complexity but compromised the performance from MAP. The authors in [6] proposed a unitary approximate message passing (UAMP) that shows a superior performance. In [7], the author proposed a variational Bayes (VB) detector that can obtain the global optimum of the approximate message passing (AMP) regardless whether the graph is loopy or not. In [8], the authors considered a cross domain signal detection and exchange extrinsic soft information iteratively in different domains, which can also be exploited in our proposed system model. Other non-linear detectors also have been proposed by authors in [9] [10] [11] [12]. Linear receivers such as LMMSE in [13] generally has a simpler algorithm. In [14], the authors applied block circular matrices properties to simplify matrix inversion. The authors in [15] and [16] applied the successive and parallel interference cancellation (SIC, PIC) for the MMSE signal detection.

Recently the authors in [17] proposed a maximum ratio combining (MRC) or matched filtering for a zero-padded OTFS system that shows a better performance than MPA. The channel structure for zero-padded OTFS inspired us to further investigate the layered SIC-MMSE detection similar to that in the superposition coding structure such as Vertical-Bell laboratories layered spacetime (VBLAST) scheme or multi-user CDMA, where the multiple copies of the superimposed signals in the receive antennas will be combined to form a better estimate at the receiver [18]. Unlike the work that has been addressed for LMMSE OTFS detection in the literature, our channel-tailored MMSE will be performed on each sub-channel and the receive signals will be processed across delay-Doppler and time domains. In the following section, we will first discuss SIC operation for each targeted layer's signal superimposed by, in our case, resolved time-variant multipaths. We will use linear filter MMSE to combine the signals, suppress the interference and eventually maximize the signal-to-noise-plus-interference ratio (SINR).

II. OTFS SYSTEM MODEL

In this paper, we will consider a zero-padded uncoded OTFS (ZP-OTFS) modulation system as explained in [17]. The RAKE receiver design in [17] motivates us to experi-

ment on MMSE filtering that has the inherent de-correlation process and can balance the disadvantages for matched filtering and zero forcing detectors by maximizing the signal-to-interference-plus-noise ratios (SINR), which potentially can achieve a better performance.

A. Transmitter

Let $x(k, l)$, $k \in [1, N]$, $l \in [1, M]$ denote the $M \times N$ Q-QAM modulated data symbols that are arranged on the delay-Doppler domain, where N and M are the number of time slots and sub-carriers per OTFS frame, respectively. The data in the delay-Doppler (DD) domain will be first transformed into the time-frequency (TF) domain by using the inverse symplectic fast Fourier transform (ISFFT) given as [1]

$$X(n, m) = \frac{1}{\sqrt{NM}} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x(k, l) e^{j2\pi(\frac{nk}{N} - \frac{ml}{M})}, \quad (1)$$

which can also be represented in a matrix format as

$$\mathbf{X}_{\text{TF}} = \mathbf{F}_M \cdot \mathbf{X}_{\text{DD}} \cdot \mathbf{F}_N^\dagger, \quad (2)$$

where $\mathbf{X}_{\text{TF}} \in \mathbb{C}^{M \times N}$ denotes the symbol matrix in the TF domain and $\mathbf{X}_{\text{DD}} \in \mathbb{C}^{M \times N}$ denotes the Q-QAM symbol matrix in DD domain. Then the symbols in the TF domain will be transformed into time domain through a multicarrier modulator that is also referred to as Heisenberg transform [1]

$$s(k) = \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} X(n, m) g_{\text{tx}}\left(\frac{kT}{M} - nT\right) e^{j2\pi m \Delta f (\frac{kT}{M} - nT)}, \quad (3)$$

here we have the sampled $g_{\text{tx}}(t)$ with sampling rate $\frac{T}{M}$, which in a matrix format can be expressed as

$$\mathbf{s} = \text{vec}(\mathbf{G}_{\text{tx}} \cdot (\mathbf{F}_M^\dagger \cdot \mathbf{X}_{\text{TF}})), \quad (4)$$

where $\mathbf{G}_{\text{tx}}$ is the matrix representation of the pulse shaping filter $g_{\text{tx}}(t)$ at the transmitter. In this paper, only rectangular pulse will be considered on both transmitter and receiver sides as ideal pulse shape cannot be realized in practice due to Heisenberg uncertainty [19]. This results in $\mathbf{G}_{\text{tx}}$ being an identity matrix, i.e., $\mathbf{G}_{\text{tx}} = \mathbf{I}_M^1$. The time domain signals are generated by an M -point FFT and an N -point IFFT of $\mathbf{X}_{\text{DD}}$ from ISFFT, followed by an M -point IFFT from Heisenberg transform. Therefore, the time domain signals can be equivalently generated by an N -point IFFT for each row of $\mathbf{X}_{\text{DD}}$, and this is also known as the inverse discrete Zak transform. Therefore, the OTFS modulated signals in time domain on the transmitter side, in the vector format, can be simplified to

$$\mathbf{s} = \text{vec}(\mathbf{X}_{\text{DD}} \cdot \mathbf{F}_N^\dagger). \quad (5)$$

¹Rectangular pulses will have high out of band emission and severe interference with practical receivers. For simplicity, we ignore these effects here and address them in a separate paper.

B. Channel

The presence of transmitter, receiver and/or scatter mobility causes wireless channels to become time selective due to the Doppler effect. Together with multi-path propagation and frequency selectivity, the channel transforms from any linear time-invariant system into a linear time-variant (LTV) system. A physically meaningful representation of the LTV channel is based on the time delays and Doppler frequency shifts. Let P denote the number of discrete paths in the delay-Doppler domain. The channel model has a sparse representation in the delay-Doppler domain given as [20]

$$h(\tau, \nu) = \sum_{i=1}^P h_i \delta(\tau - \tau_i) \delta(\nu - \nu_i), \quad (6)$$

where τ_i , ν_i and h_i are the delay, Doppler shift and attenuation factor for the i -th path, respectively [20]. In addition, the continuous impulse response of the channel in the delay-time domain is related to Equation (6) by

$$h(\tau, t) = \int_{\nu} h(\tau, \nu) e^{j2\pi \nu(t-\tau)} d\nu. \quad (7)$$

The delay and Doppler can be evaluated at $\tau_i = \ell_i \frac{T}{M}$, $\nu_i = \kappa_i \frac{\Delta f}{N}$, where ℓ_i and κ_i are the delay taps and doppler taps for the i -th path and they are not necessarily integers. T and Δf are symbol period and sub-carrier frequency, respectively. Substituting (6) we then have

$$h(\ell, t) = \sum_{i=1}^P h_i e^{j2\pi \kappa_i \frac{\Delta f}{N} (t - \ell \frac{T}{M})} \delta(\ell - \ell_i) \delta(\kappa - \kappa_i). \quad (8)$$

The discrete baseband equivalent channel seen by the receiver will be the sinc reconstructed impulse response having integer multiples l of delay resolution $\frac{T}{M}$ on delay τ , with discrete sampling intervals $T_s = n \frac{T}{M}$ on time t . Thus, the discrete baseband equivalent delay-time channel model is given as

$$h_e(l, n) = h(l, t)|_{t=n \frac{T}{M}} \text{sinc}(l - \ell), \quad (9)$$

where $\text{sinc}(x) = \sin(\pi x)/(\pi x)$. It is also assumed that the fractional delay taps ℓ can be modelled as integer delay taps for ultra-wide band systems². In this case, $\text{sinc}(l - \ell)$ will be evaluated to 1 when $\ell = l \in \mathbb{Z}$. Then from Equation (9) we will have the integer delay tap baseband equivalent channel

$$h_e(l, n) = \sum_{i=1}^P h_i z^{\kappa_i(n-l)}, \quad (10)$$

where $z = e^{j2\pi \frac{\Delta f}{NM}}$. Note that κ_i are the Doppler taps of the i -th path and they are not necessarily integers. Assumptions of integer Doppler taps, on the other hand, will introduce performance loss [17], hence we will consider fractional Doppler taps throughout this paper.

²Existing practical 4G and 5G systems always have fractional delays.

C. Receiver

At the receiver side, the receive signal $r(t)$ can be modeled as a convolution with the channel response $h(\tau, t)$ described by

$$r(t) = \int_0^{\tau_{max}} h(\tau, t) s(t - \tau) d\tau. \quad (11)$$

Then, $r(t)$ is first sent into a multicarrier demodulator with a pulse shaping filter to obtain $Y(t, f)$ [1]

$$Y(t, f) = \int g_{rx}^*(t - nT) r(t) e^{-j2\pi f(t - nT)} dt. \quad (12)$$

Then the signal in the TF domain is sampled at $t = nT, f = m\Delta f$

$$Y(n, m) = Y(t, f)|_{t=nT, f=m\Delta f}. \quad (13)$$

This is followed by symplectic fast Fourier transform (SFFT) to convert TF signals into DD domain.

$$Y(k, l) = \frac{1}{\sqrt{NM}} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} Y(n, m) e^{-j2\pi(\frac{nk}{N} - \frac{ml}{M})}. \quad (14)$$

D. Input-Output Relations in Different Domains

1) *Delay Doppler Domain*: Let $\mathbf{x}_m$ denote the m -th row vector of $\mathbf{X}_{DD}$ and $\mathbf{y}_m$ denote the m -th row vector of $\mathbf{Y}_{DD}$. The IOR in the DD domain for each transmit and receive vector $\mathbf{x}_m$ and $\mathbf{y}_m$ is given by [17]

$$\mathbf{y}_m = \sum_{l \in \mathcal{L}} \boldsymbol{\nu}_{m,l} \otimes \mathbf{x}_{m-l}, \quad (15)$$

where l is the integer delay tap and $\boldsymbol{\nu}_{m,l}$ is the channel response vector and it is given by

$$\boldsymbol{\nu}_{m,l}(k) = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} \sum_{i=1}^P h_i z^{\kappa_i(m-l)} e^{\frac{j2\pi(\kappa_i - k)n}{N}}, \quad (16)$$

for integer delays and fractional dopplers. More explicitly, the DD domain discrete channel can be holistically expressed as

$$\mathbf{y} = \mathbf{H} \cdot \mathbf{x} + \mathbf{w}, \quad (17)$$

with each sub-matrix $\mathbf{K}_{m,l} \in \mathbb{C}^{N \times N}$ in $\mathbf{H}$ being a banded matrix representing the convolution relationship in Equation (15). Fig. 1 shows the IOR in the DD domain. Each transmit signal vector $\mathbf{x}_m$ consists of N symbols that are located in the same delay slot m . And each sub-matrix $\mathbf{K}_{m,l}$ is given as

$$\mathbf{K}_{m,l} = \begin{bmatrix} \boldsymbol{\nu}_{m,l}(0) & \boldsymbol{\nu}_{m,l}(N-1) & \cdots & \boldsymbol{\nu}_{m,l}(1) \\ \boldsymbol{\nu}_{m,l}(1) & \boldsymbol{\nu}_{m,l}(0) & \cdots & \boldsymbol{\nu}_{m,l}(2) \\ \vdots & \ddots & \ddots & \vdots \\ \boldsymbol{\nu}_{m,l}(N-1) & \boldsymbol{\nu}_{m,l}(N-2) & \cdots & \boldsymbol{\nu}_{m,l}(0) \end{bmatrix} \quad (18)$$

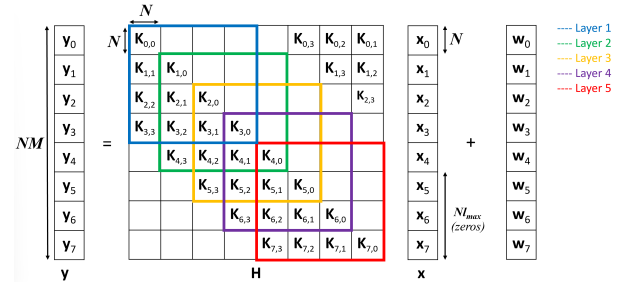


Fig. 1. Delay-Doppler domain input-output relation for $N = M = 8$ and $l_{max} = 3$

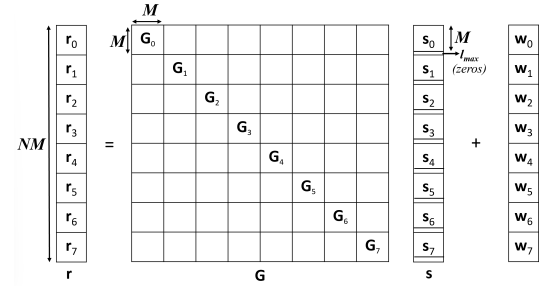


Fig. 2. Time domain input-output relation for $N = M = 8$ and $l_{max} = 3$

2) *Time Domain*: Similar to DD domain operation, we will also inspect the transmit signals in the time domain by groups and provide the channel response for each group from [17]. Let $\mathbf{s}_n$ denote symbol vector transmitted in the n -th time slot. And let $\mathbf{r}_n$ denote the receive signals in the n -th time slot. The channel in the time domain for all time slots signals can be expressed as

$$\mathbf{r} = \mathbf{G} \cdot \mathbf{s} + \tilde{\mathbf{w}}. \quad (19)$$

The channel response vector in delay-time domain is

$$\tilde{\boldsymbol{\nu}}_{m,l}(n) = \sum_{i=1}^P h_i z^{\kappa_i(m-l)} e^{\frac{j2\pi\kappa_i n}{N}}. \quad (20)$$

And each sub-matrix of $\mathbf{G}$ in the time domain is related to (20) as

$$\mathbf{G}_n(m, m-l) = \tilde{\boldsymbol{\nu}}_{m,l}(n). \quad (21)$$

Because Doppler cannot be resolved in this domain, the fractional Doppler does not complicate the structure of $\mathbf{G}$, and in return low complexity detection such as MMSE can be realized in this domain for each time slot signals. Fig. 2 shows the time domain IOR for all time domain signals. In addition, if we inspect the internal structure of each sub-matrix $\mathbf{G}_n$ that represents the channel response for each transmit signal vector $\mathbf{s}_n$ of size M , the IOR for each time slot signal $\mathbf{s}_n$ and $\mathbf{r}_n$ will form the structure in Fig. 3, where each s_{nm} is the $(m+1)$ -th single element in $\mathbf{s}_n$ and together they comprise $\mathbf{s}_n = [s_{n0}, s_{n1}, \dots, s_{nm}]$. Now given the IORs in the time domain for all signals and the individual n -th signal vectors, instead of detecting the N symbols in each time slot at one time from one MMSE on the large sub-channel size in the DD domain, we will detect each symbol s_{nm} one by one in $\mathbf{s}_n$ in

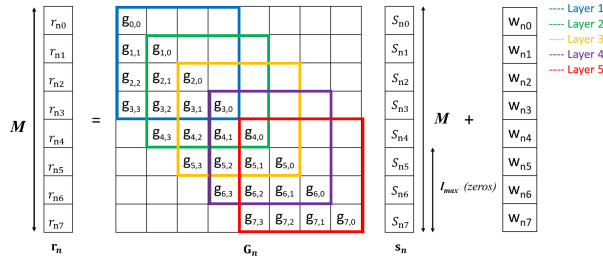


Fig. 3. Input-output relation for each transmit vector $\mathbf{s}_n$ and receive vector $\mathbf{r}_n$

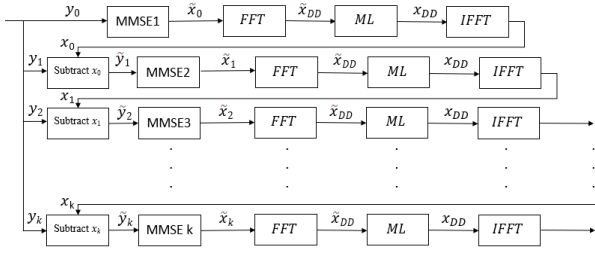


Fig. 4. SIC-MMSE receiver for OTFS modulation in time domain

the time domain based on the IOR given in Fig. 3. And this process will be repeated for the other $N - 1$ vectors. Details of the time domain detection will be discussed in the following section.

III. ITERATIVE MMSE FOR OTFS

In this section, we will introduce the MMSE detection with successive interference cancellation (SIC) for the zero-padded OTFS system. The transmit signals will be detected on the basis of layers or levels. The noise plus interference from other layers, if any, will be suppressed through MMSE filtering and the SINR will be maximized for current targeted layer. From SIC, the interference caused by previous time slot symbols or layers will be subtracted sequentially from the current iteration estimates, and the interference caused by the following time slots will be canceled using the estimates in the previous iteration.

A. MMSE Detection with Successive Interference Cancellation in the Delay-Doppler Domain

Let $\mathbf{x}_m \in \mathbb{C}^{N \times 1}$ denote the to-be-detected symbol vector for layer $m + 1$ and $\mathbf{H}_{m:n} \in \mathbb{C}^{N(l_{max}+1) \times N(l_{max}+1)}$ denote the square sub-channel of $\mathbf{H}$ corresponding to the effective channel for the $(m + 1)$ -th layer signals and its interference, which is reflected in different colors in Fig. 1. l_{max} here represents the largest delay spread tap. Then the receive signal vector for layer $m + 1$ are $\mathbf{y}_{m:m+l_{max}}$ that consists of the targeted layer symbols, its interference, and noise. The details of MMSE and SIC operation in the DD domain is discussed in the following by taking the example in Fig. 1.

1) *Layer 1*: There is no interference from previous time slot exerted on the first-layer transmit symbol vector $\mathbf{x}_0$ due to the insertion of zero paddings. Therefore, we directly perform

MMSE filtering on the receive signal vector $\mathbf{y}_{0:l_{max}}$, which will be $\mathbf{y}_{0:3}$ in our example from Fig. 1. The MMSE filter coefficients for this layer are given by

$$\mathbf{w}_0^{MMSE} = \mathbf{h}_0^\dagger [\mathbf{H}_{0:3} \mathbf{H}_{0:3}^\dagger + \sigma^2 \mathbf{I}]^{-1}, \quad (22)$$

where $\mathbf{h}_0 = [\mathbf{K}_{0,0}, \mathbf{K}_{1,1}, \mathbf{K}_{2,2}, \mathbf{K}_{3,3}]^T \in \mathbb{C}^{N(l_{max}+1) \times N}$, and $\sigma^2 = \sigma_n^2/E_s$. And then the filtered layer 1 signal vector can be expressed as

$$\tilde{\mathbf{x}}_0^{est} = \mathbf{w}_0^{MMSE} \mathbf{y}_{0:3}, \quad (23)$$

followed by making hard decisions from the maximum likelihood criterion

$$\hat{\mathbf{x}}_0(n) = \arg \min_{a_j \in \mathcal{Q}} |a_j - \tilde{\mathbf{x}}_0^{est}(n)|, \quad (24)$$

where a_j is signal from the QAM alphabet $\mathcal{Q}$ with $j = 1, \dots, |\mathcal{Q}|$ and $n = 0, \dots, N - 1$.

2) *Layer 2*: Given the estimated signal vector $\hat{\mathbf{x}}_0$ from the first layer, we will first subtract them from the receive signal vector $\mathbf{y}_{1:4}$ by

$$\tilde{\mathbf{y}}_{1:4} = \mathbf{y}_{1:4} - \mathbf{h}_0 \hat{\mathbf{x}}_0. \quad (25)$$

Then we have the MMSE filter coefficients for this second layer calculated as

$$\mathbf{w}_1^{MMSE} = \mathbf{h}_1^\dagger [\mathbf{H}_{1:4} \mathbf{H}_{1:4}^\dagger + \sigma^2 \mathbf{I}]^{-1}, \quad (26)$$

where $\mathbf{h}_1 = [\mathbf{K}_{1,0}, \mathbf{K}_{2,1}, \mathbf{K}_{3,2}, \mathbf{K}_{4,3}]^T \in \mathbb{C}^{N(l_{max}+1) \times N}$. And the estimated layer 2 signal vector will be

$$\tilde{\mathbf{x}}_1^{est} = \mathbf{w}_1^{MMSE} \tilde{\mathbf{y}}_{1:4}, \quad (27)$$

followed by making hard decisions

$$\hat{\mathbf{x}}_1(n) = \arg \min_{a_j \in \mathcal{Q}} |a_j - \tilde{\mathbf{x}}_1^{est}(n)|. \quad (28)$$

3) *Layer k+l*: Generally in the $(k + 1)$ -th layer, the receive signal vectors after interference cancellation in the first iteration are given by

$$\tilde{\mathbf{y}}_{k:k+l_{max}} = \mathbf{y}_{k:k+l_{max}} - \sum_{i=0}^{k-1} \mathbf{h}_i \hat{\mathbf{x}}_i. \quad (29)$$

And the MMSE filter coefficients for $(k + 1)$ -th layer is

$$\mathbf{w}_k^{MMSE} = \mathbf{h}_k^\dagger [\mathbf{H}_{k:k+l_{max}} \mathbf{H}_{k:k+l_{max}}^\dagger + \sigma^2 \mathbf{I}]^{-1}. \quad (30)$$

This process is repeated until reaching the last layer when our first iteration detection is completed.

B. Iterative SIC-MMSE Detection

The performance of SIC-MMSE can be further improved by extending the previous detection process with iterations. For each layer k , the interference caused by the latter layers $k + 1, k + 2, \dots, k + l_{max}$ can be cancelled by using the estimates in the previous iteration and the interference from the previous layers $0, 1, \dots, k - 1$ can be cancelled by using the estimates in the current iteration. In other words, after

performing the interference cancellation, the receive signals in the i -th iteration can be calculated as

$$\tilde{\mathbf{y}}_{k:k+l_{max}}^i = \mathbf{y}_{k:k+l_{max}}^i - \sum_{m=0}^{k-1} \mathbf{h}_m \hat{\mathbf{x}}_m^i - \sum_{m=k+1}^{k+l_{max}} \mathbf{h}_m \hat{\mathbf{x}}_m^{i-1}, \quad (31)$$

where $\hat{\mathbf{x}}_m^i$ are the hard estimates in the i -th iteration for symbol vector $\hat{\mathbf{x}}_m$, and $\hat{\mathbf{x}}_m^{i-1}$ are the hard estimates from $(i-1)$ -th iteration. The MMSE filtering coefficients therefore can be obtained by

$$\mathbf{w}_k^{MMSE} = \mathbf{h}_k^\dagger [\mathbf{h}_k \mathbf{h}_k^\dagger + \sigma^2 \mathbf{I}]^{-1}. \quad (32)$$

Note that we assume the perfect interference cancellation can be achieved without considering residual interference power. The complexity of performing MMSE is quite high due to matrix inversion in (30) and (32), of order $O(MN^3L^3)$, where $L = l_{max}$. By changing detection from DD domain to time domain, a lower complexity order of $O(MNL^3)$ can be achieved instead.

C. Low Complexity MMSE in the Time Domain

The complexity of performing MMSE detection mostly arises from calculating the matrix inversion of $\mathbf{A} = \mathbf{H}_{k:k+l_{max}} \mathbf{H}_{k:k+l_{max}}^\dagger + \sigma^2 \mathbf{I}$ in (30), and this requires a computational complexity of $O(MN^3L^3)$ in the DD domain. However, the time domain channel has a simpler structure that is not impacted by Doppler effect as Doppler cannot be resolved in this domain. Instead of detecting signals in groups from one MMSE operation, which has a larger sub-channel size, we will detect symbols one by one in the time domain and transform them into DD domain to make hard decisions. The effective channel matrix for detecting each symbol only has the size of the largest delay spread tap $L = l_{max}$ which typically is less than 20 for EVA channel in our simulations. And therefore the overall complexity for obtaining the matrix inversion can be reduced to $O(MNL^3)$.

1) *SIC-MMSE Detection in Time Domain*: Let s_{nm} denote the $(m+1)$ -th layer symbol in the n -th time slot and r_{nm} denote the receive signals in the time domain reflected in Fig. 3. The time domain receive signals $\tilde{\mathbf{r}}$ after interference cancellation corresponding to $(k+1)$ -th layer transmit signal s_{nm} are given as

$$\tilde{\mathbf{r}}_n = \mathbf{r}_n - \sum_{m=0}^{k-1} \mathbf{g}_m \hat{s}_{nm}^i - \sum_{m=k+1}^{k+l_{max}} \mathbf{g}_m \hat{s}_{nm}^{i-1}, \quad (33)$$

where $\mathbf{r}_n$ is the receive symbols from k to $(k+l_{max})$ in the n -th time slot, $\mathbf{g}_m$ is the channel vector in column $(m+1)$ in the sub-matrix $\mathbf{G}_n$. $\hat{s}_{nm}^i$ are the symbols that have been updated in the i -th iteration, and $\hat{s}_{nm}^{i-1}$ are the symbols that were detected in the last $(i-1)$ -th iteration. The MMSE filter coefficients for estimating signal s_{nm} is then

$$\mathbf{w}_{nm}^{MMSE} = \mathbf{g}^\dagger [\mathbf{H}_e \mathbf{H}_e^\dagger + \sigma^2 \mathbf{I}]^{-1}, \quad (34)$$

where $\mathbf{H}_e$ is the square effective sub-channel for $(k+1)$ -th layer symbol reflected in different colors shown in Fig. 3,

and $\mathbf{g}$ is the first column in $\mathbf{H}_e$. Compared with the size of effective sub-channel matrix in DD domain $N(l_{max}+1) \times N(l_{max}+1)$, the sub-channel matrix size in the time domain is only $(l_{max}+1) \times (l_{max}+1)$. Now the $(k+1)$ -th layer symbol in the n -th time slot after MMSE is then

$$\tilde{s}_{nm} = \mathbf{w}_{nm}^{MMSE} \tilde{\mathbf{r}}_n. \quad (35)$$

The rest of the $(k+1)$ -th layer symbol in all the other time slots $N-1$ will be evaluated in the same way and we will eventually have a length N estimated signal vector $\tilde{\mathbf{s}}_m$. Then the estimated symbol vector will be transformed into DD domain through an N -point FFT

$$\tilde{\mathbf{x}}_{DD,m} = \mathbf{F}_N \tilde{\mathbf{s}}_m. \quad (36)$$

And the hard decisions are made by the maximum likelihood criterion

$$\hat{x}_{DD,m}(n) = \arg \min_{a_j \in \mathcal{Q}} |a_j - \tilde{x}_{DD,m}(n)|. \quad (37)$$

Then the hard decision symbol vector will be transformed back into time domain through an N -point IFFT

$$\hat{\mathbf{s}}_m = \mathbf{F}_N^\dagger \hat{\mathbf{x}}_{DD,m}. \quad (38)$$

The vector $\hat{\mathbf{s}}_m$ is the hard estimates in the time domain for all the N signals in the m -th delay slot, and they will be subtracted as interference on the $(m+1)$ -th targeted receive signals when the next round of detection is performed. The time domain SIC-MMSE operation is also illustrated in Fig. 4.

IV. SIMULATION RESULTS

In this section, we will provide simulation results for different OTFS frame sizes and QAM modulation orders. In line with current communication standards, the sub-carrier spacing Δf is taken as 15 kHz and carrier frequency being 4 GHz. The wireless channel is generated according to the standard EVA channel model with a UE speed at 120 km/h. The maximum integer delay tap is $l_{max} = 19$ and the maximum fractional Doppler tap is $k_{max} = 3.79$ for $M = 512$ and $N = 128$. The Doppler taps for the i -th path will be generated from a uniform distribution $U(-k_{max}, k_{max})$. A perfect channel state information is assumed at the receiver. BER plots are shown for the proposed detection in an uncoded zero-padded OTFS system. Fig. 5 shows the BER plot for our proposed SIC-MMSE, compared with MRC and MPA detectors. A maximum of 10 iterations is allowed for these iterative detectors. It can be found out that our proposed SIC-MMSE detection outperforms the MRC detector without having to perform initial time-frequency estimates. When E_s/σ^2 (where E_s is the symbol energy and σ^2 is noise power) approaches small values, the performance of our proposed SIC-MMSE is similar to the MRC or matched filter. This is sensible as low E_s/σ^2 accounts for a high noise-power-to-symbol-energy ratio which makes MMSE filter coefficient in Equation (42) mainly affected by $\mathbf{h}^\dagger$ that is essentially a matched filter. When E_s/σ^2 is high, Equation (42) can be approximated by $\mathbf{h}^\dagger (\mathbf{H}\mathbf{H}^\dagger)^{-1}$, and

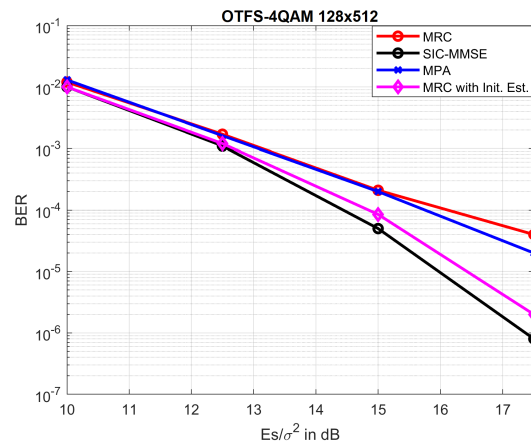


Fig. 5. Uncoded 4-QAM BER plot: MRC vs MPA vs SIC-MMSE

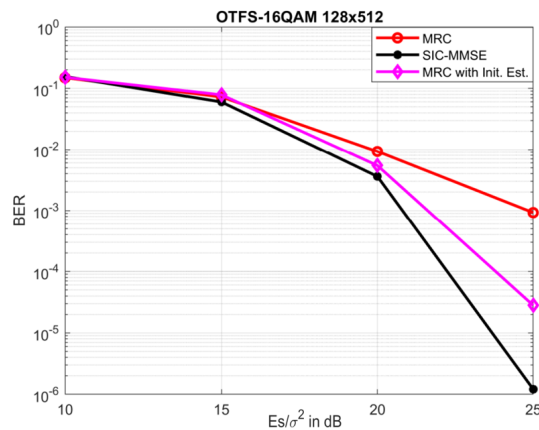


Fig. 6. Uncoded 16-QAM BER plot: MRC vs SIC-MMSE

MMSE will become a zero forcing detector. With a single-tap equalizer added in the time-frequency domain for MRC, it can generate a drastic improvement; however, our proposed SIC-MMSE still shows a better performance especially at high E_s/σ^2 even though our proposed SIC-MMSE does not need time-frequency initial estimates. Fig. 6 shows the BER plot for 16-QAM modulation with a maximum 15 iterations allowed. Similar to 4-QAM performance, the proposed SIC-MMSE for 16-QAM shows an approximately equal BER at low E_s/σ^2 , and surpasses the other two methods at high E_s/σ^2 .

V. CONCLUSION

In this article, we investigated the performance of the zero-padded OTFS modulation system. We proposed a successive interference cancellation based iterative MMSE detection for zero-padded OTFS system. A low complexity detection in the time domain was also proposed. The performance of proposed MMSE detection can outperform the state-of-the-art detectors such as MPA and RAKE receivers.

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